

UWAF T Onboarding Challenge

Design Objectives:

Create a mount for a rotary switch that can hold the switch safely upon drywall, protect it from external forces, and protect its electrical components such that it can connect to a greater system.

Required Materials:

4 x M3 Brass Heat Set Inserts, 5.7 mm length

4 x M3 Brass Pan Head Phillips Screws, 8 mm length

4 x #6 drywall screws, 1 5/8 inches length

Around 700 mm x 700 mm of 1.5 mm thickness low carbon steel

Given and Implied Constraints:

Constraint: Mount should cover the open side panels but leave the top switch exposed

Solution: Mount will include four 1.5 mm thick walls which cover the sides, and a 56.25 mm hole on the top face which exposes the top switch.

Constraint: The mounting holes on the switch can also be used for the mount

Solution: Mount will include four 5.5 mm diameter holes on the top face which enables us to use the mounting holes on the switch.

Constraint: The mount should securely mount the switch to a wall (can assume drywall)

Solution: The mount will include four #6 drywall screws which secure the mount's base and the switch to the drywall.

Constraint: The switch must be detachable from the mount.

Solution: The switch will be separate from the mount, and secured to the mount through four M3 screws on the top face which allow the switch to be easily removed to make adjustments or changes to its individual parts.

Constraint: The switch must be secure within the mount.

Solution: Four brass M3 heat set inserts will be melted into the polycarbonate mounting holes, and four brass m3 screws will be screwed into the inserts to secure the switch to the top of the mount.

Constraint: Cables must be able to exit the switch and attach to the mainframe.

Solution: A 5 mm gap is given between the switch and the mount walls to accommodate up to AWG 22 cables.

Constraint: Switch mount should not break due to high amounts of stress applied to the switch.

Solution: See the Load-Bearing Calculations section.

Constraint: Switch mount should not fall out if pulled with a large amount of force.

Solution: See the Load-Bearing Calculations section.

Constraint: Switch should survive mildly dangerous conditions, like water, rain, or snow falling on the switch.

Solution: The mount covers the maximal amount of surface area around the switch and protects the switch from water leaks which could otherwise compromise the electronic parts within.

Constraint: No part of the mount can be a safety hazard due to its geometry.

Solution: The edges of the mount are filleted by welds to prevent cuts from hitting one's hand on a sharp edge.

Design Strategy:

The design of the mount distributes the force applied on the switch to two sets of screws: one attaching the switch to the mount and another attaching the mount to the drywall. Using two sets of screws gives the mount and switch more stability and screw strength. On the other hand, sticking long screws through the entire clearance hole of the screw will cause it to bear too much torque when force is applied to the switch and makes removing the switch a tedious and lengthy process.

Load-Bearing Calculations:

4 screws distribute force evenly to all screws

Maximum force tolerance: 100 N

Shear stress: $F/A = (25 \text{ N}) / (15 \times 10^{-3} \text{ m})(2.98 \times 10^{-3}) = 5.6 \times 10^6 \text{ Pa} \approx 5.6 \text{ MPa}$

Steel shear stress = 344 MPa

Maximum pull out force:

$$F_{\text{pull}} = \pi * (2.98 * 10^{-3}) * (5.7 * 10^{-3}) * (39.9 \text{ MPa}) = 2129 \text{ N}$$

These basic mechanical calculations prove that the mount will not break or be torn from the wall even if significant manual force is applied towards the mount or switch.

Manufacturing Process:

1. First, solder the M3 heat set inserts into the 5.5 diameter holes in the switch.
2. Cut a 124.93 mm x 124.93 mm sheet of low carbon steel. Call this large sheet A1.
3. Then, cut a 87.93 mm x 87.93 mm sheet from the middle of sheet A1. Label this sheet A2.
4. Make 4 corner fillets on sheet A2 of radius 11.5 mm.
5. Punch 4 holes of diameter 5.5 mm which match the positions on the switch.
6. Punch a large hole in the center of sheet A2 of diameter 56.25 mm.
7. Punch 4 holes of diameter 3.7 mm on sheet A1 according to the mechanical diagram.
8. Cut a 62.31 mm by 351.72 mm sheet of low carbon steel. Call this large sheet A3.
9. Bend sheet A3 three times with radius 11.5 mm as according to the mechanical diagram.
10. Weld the last edge of sheet A3 to the first edge.
11. Weld sheet A3 overtop sheet A1.
12. Weld sheet A2 overtop sheet A3.
13. Place the switch inside of the mount, then add screws to the switch and to the drywall.

Cost Efficiency and Potential Improvements:

Current cost of materials:

24" x 24" Low Carbon Steel Sheet from McMaster Carr - 0.048" thick - \$39.23

4 x M3 Brass Heat Set Inserts - $\$0.22 \times 4 = \0.88

4 x M3 Brass Pan Head Phillips Screws, 8 mm length - $\$0.15 \times 4 = \0.60

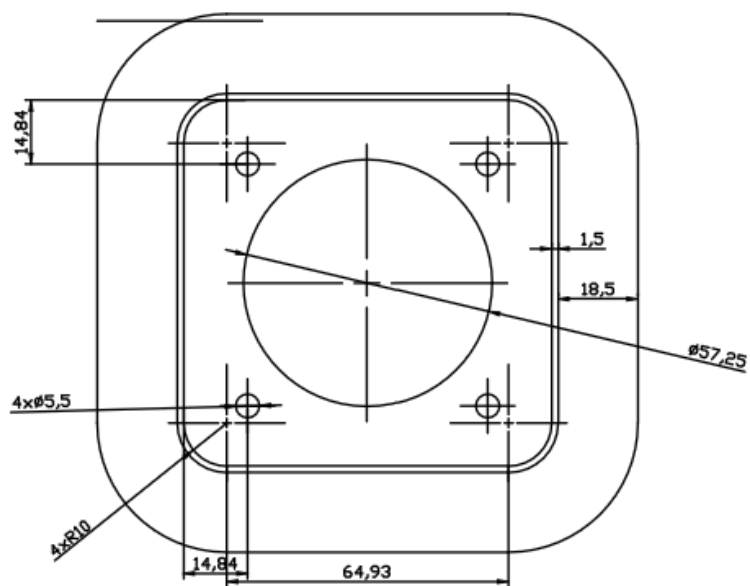
4 x #6 drywall screws, 1 5/8 inches length - $\$0.06 \times 4 = \0.24

Total cost: \$40.95

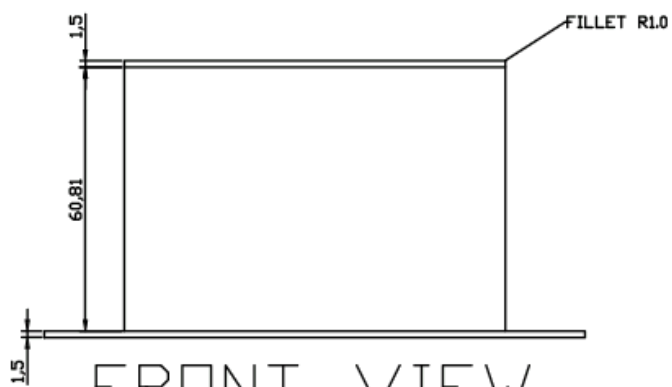
Potential improvements include:

- Reducing the thickness of the steel material.
- Exchanging steel for a cheaper and more easily manufacturable material.
- Cutting surface area from the mount to reduce cost.

Diagram and Dimensioning:



TOP VIEW



FRONT VIEW

